

Total No. of Printed Pages:2

SUBJECT CODE NO: - RNP-8020**FACULTY OF SCIENCE AND TECHNOLOGY****S.E (Civil Engg.) (NEP) Sem - III****Examination November / December 2025 (To be held in February-2026)****Strength of Material****[Time: 2 Hours]****[Max. Marks:30]**

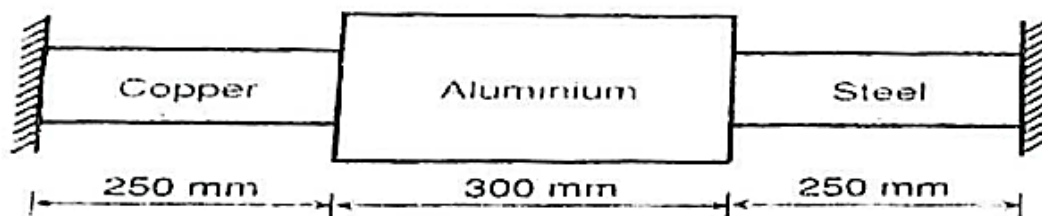
Please check whether you have got the right question paper.

N. B

1. Part A: Question No 1 is compulsory.
2. Attempt any TWO questions from remaining Part B: Q.2 to Q.5.

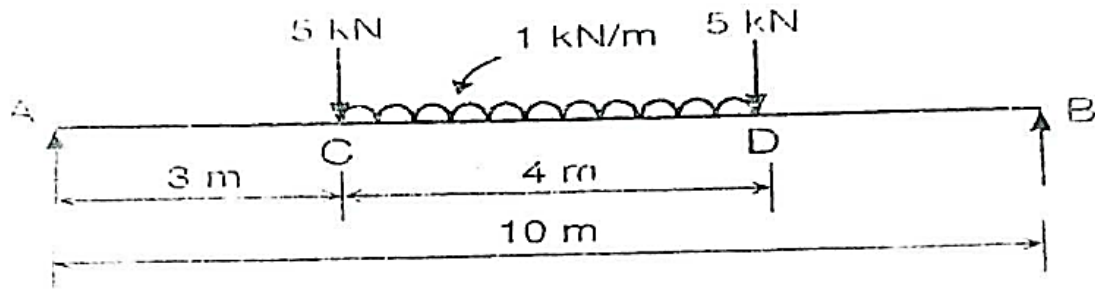
Part A**Q1** Solve any 5 following questions**10**

- a) Define normal stress and write its mathematical expression.
- b) State Hooke's law.
- c) What is principal stress?
- d) Define bending moment.
- e) Write the torsion formula for circular shafts.
- f) Define slenderness ratio.
- g) What is Saint Venant's principle?

Part B**Q.2** Composite bar is as shown in fig. If the temperature of the system is raised by 80°C . Calculate the stresses developed in three parts. **10**Take: $E_c = 100 \text{ GPa}$, $\alpha_c = 10 \times 10^{-6} / ^{\circ}\text{C}$, $d_c = 60 \text{ mm}$ $E_a = 80 \text{ GPa}$, $\alpha_a = 20 \times 10^{-6} / ^{\circ}\text{C}$, $d_a = 85 \text{ mm}$ $E_s = 200 \text{ GPa}$, $\alpha_s = 12 \times 10^{-6} / ^{\circ}\text{C}$, $d_s = 60 \text{ mm}$ 

Q.3 Draw SFD and BMD of given beam.

10



Q.4 T Section with flange ($200 \text{ mm} \times 50 \text{ mm}$) and web ($50 \text{ mm} \times 125 \text{ mm}$), is used as a cantilever beam of 3 m span subjected to a u.d.l. of intensity 20 kN/m over its full span. Determine the maximum stresses in the beam.

10

Q.5 Analyse the given truss by joint method. Also draw the nature of forces in each member.

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